D1.2 Protein synthesis

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-d1-2.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 625–636 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

Transcription

D1.2.12 Direction

pp. 625–626

RNA polymerase reads the template strand 3′ to 5′ and builds RNA **5′ to 3′**, adding nucleotides to the free 3′ -OH. Ribosomes read mRNA **5′ to 3′**, building the polypeptide from its amine end to its carboxyl end.

D1.2.13 Starting at the promoter

pp. 626–627

General **transcription factors** bind the **promoter** just before a gene; RNA polymerase then joins them. Activators bound to enhancers can loop round and raise the rate. Without the right factors, the gene is not transcribed. RNA polymerase itself is not a transcription factor.

D1.2.14 Non-coding DNA

pp. 627–628

Only about 1.5% of human DNA codes for polypeptides. Non-coding sequences include regulators of gene expression (promoters, enhancers), **introns**, **telomeres** that protect chromosome ends during replication, and genes for **rRNA and tRNA**, whose RNA is the final product.

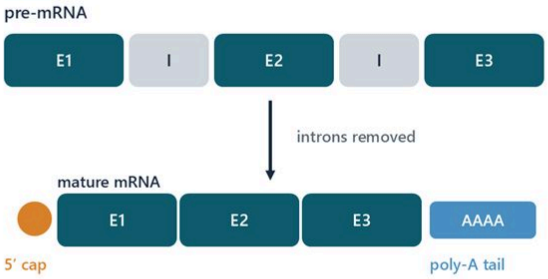
Processing

D1.2.15 Post-transcriptional modification

pp. 628–629

Processing mRNA in eukaryotes

D1.2.15



Change	Purpose
Introns out, exons spliced	Only coding sequence left
5′ cap	Protects; helps the ribosome bind
Poly-A tail	Protects; helps export

Where and why

This happens in the nucleus. Prokaryotes have no introns, so to make a human protein in bacteria an intron-free copy of the gene (made from mRNA) is used.

D1.2 Protein synthesis (HL) – Textbook pp. 628–629 9

In eukaryotes, **introns** are removed from pre-mRNA and **exons** spliced together; a **5′ cap** and a **3′ poly-A tail** are added, which protect the mRNA and help it leave the nucleus and bind ribosomes.

D1.2.16 Alternative splicing

pp. 629–630

Different combinations of exons can be kept, so one gene can give several versions of a protein. The **troponin T** gene is spliced differently in foetal and adult heart muscle.

Translation and after

D1.2.17 Starting translation

pp. 630–633

The small subunit binds the 5′ end of mRNA and moves to the start codon AUG; the initiator tRNA (methionine, anticodon UAC) pairs with it; the large subunit joins, with the initiator tRNA in the **P site**. A tRNA with the next amino acid enters the **A site**, a peptide bond forms, the ribosome moves one codon, and the empty tRNA leaves from the **E site**.

D1.2.18 Modifying polypeptides

pp. 633–635

Many polypeptides are modified before they work. **Pre-proinsulin** loses its signal sequence in the ER, giving proinsulin, which folds with three disulfide bonds; later the C-peptide is cut out, leaving **insulin**: A and B chains held by disulfide bonds.

D1.2.19 Proteasomes

pp. 635–636

Damaged or unneeded proteins are tagged with **ubiquitin** and broken down by **proteasomes**. The amino acids are reused, so a working proteome depends on constant breakdown as well as synthesis.

© 2026 Akash Parthi Bendran. Please ask before using or sharing: see whatthecell.org/about.html#permission